

Assignment: Requirement Analysis & System Design

Project: Call Center Voice Calling Platform

Background

Our company currently uses a third-party call center software solution for customer communication. Due to increasing operational requirements, scalability needs, and integration challenges, we have decided to build and maintain our own call center platform.

The platform will be deployed and managed on our own infrastructure.

The company will obtain telephone numbers from BTCL and use them for inbound and outbound customer communication.

Currently, the call center has more than 50 active agents, and the number of agents is expected to increase significantly in the future.

The platform will integrate with an existing CRM system through APIs to retrieve customer information and update call-related activities.

The goal is to create a modern, scalable, secure, and maintainable call center platform that can support future growth and advanced capabilities such as AI-powered features.

Business Objectives

The platform should:

- Support inbound and outbound calling.
- Provide a smooth customer experience.
- Allow agents to efficiently manage customer interactions.
- Allow management to monitor and control call center operations.
- Provide team and agent performance monitoring.
- Support call quality assurance processes.
- Store and manage call recordings.
- Dynamically distribute calls among agents.
- Be secure and easy to deploy.

- Scale as the organization grows.
 - Be designed to support future AI-powered capabilities.
 - Be extensible so new features can be added with minimal impact.
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Technical Context

Existing Systems

- CRM system already exists.
- Customer information can be accessed through CRM APIs.
- Telephony numbers will be provided through BTCL.
- Company will host and manage the solution on its own servers.

Preferred Technology Stack

- Backend: .NET Core
- Frontend: Angular

You may recommend additional technologies if required.

Your Task

Assume you are the System Analyst and Solution Architect assigned to this project.

You are responsible for analyzing the requirements, identifying missing information, defining the MVP scope, and proposing a scalable architecture.

Do not assume the requirements above are complete.

Part of your responsibility is to identify gaps, risks, assumptions, and questions that should be clarified before implementation begins.

Deliverables

1. Requirement Analysis Document

Prepare a detailed Requirement Analysis document including:

Business Goals

Stakeholders

Identify all stakeholders involved.

Assumptions

Clearly document assumptions.

Functional Requirements

Identify and document all functional requirements you believe are necessary.

Non-Functional Requirements

Examples:

- Performance
- Availability
- Reliability
- Security
- Monitoring
- Maintainability
- Scalability
- Disaster Recovery

Risks

Identify potential project risks.

Out-of-Scope Items

Clearly identify what should not be included in the MVP.

2. Stakeholder Question List

Prepare a list of questions that should be discussed with stakeholders before implementation begins.

Examples:

- Business questions
- Operational questions
- Security questions
- Telephony questions
- Compliance questions
- Scalability questions

The quality of questions will be heavily evaluated.

3. MVP Feature Definition

Based on your analysis:

Define the MVP Scope

Explain:

- Which features should be included
- Which features should be excluded
- Why

Your MVP should balance:

- Business value
 - Development effort
 - Risk
 - Time to market
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4. System Design Plan

Prepare a high-level system design.

Include:

Architecture Diagram

Major Components

Examples:

- Agent Portal
- Admin Portal
- Telephony Layer
- CRM Integration Layer
- Reporting Module
- Call Recording Service
- Authentication & Authorization
- Monitoring & Logging

Data Flow

Explain major workflows.

Database Design

Provide a proposed database structure and entity relationships.

API Design Approach

Explain how services communicate.

5. Scalability & Future Readiness

Design the system so it can evolve from:

Current State

- 50+ Agents

Future State

- 500+ Agents
- Multiple Teams
- Multiple Campaigns
- Higher call volume

Explain:

- Scalability strategy
- High availability strategy

- Load balancing strategy
 - Monitoring strategy
 - Deployment strategy
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6. AI Readiness Strategy

The company plans to introduce AI capabilities in future phases.

Explain how your architecture will support future features such as:

- AI Call Summary
- AI Call Transcription
- AI QA Review
- AI Agent Assistance
- AI-Based Call Routing

No implementation is required.

Focus on architectural readiness.

7. Deployment Strategy

Propose a deployment approach for:

- Development Environment
- Staging Environment
- Production Environment

Include:

- CI/CD approach
 - Rollback strategy
 - Monitoring
 - Backup strategy
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Deliverable Format

Submit a single document containing:

1. Requirement Analysis
2. Stakeholder Questions
3. MVP Definition
4. System Design
5. Database Design
6. Scalability Plan
7. AI Readiness Plan
8. Deployment Strategy

Time Limit

3 Days

Expected Experience Level

Software Engineer / System Analyst (4+ Years Experience)

Preferred Deliverable Quality

The document should be detailed enough that a development team could use it as the starting point for implementation planning.